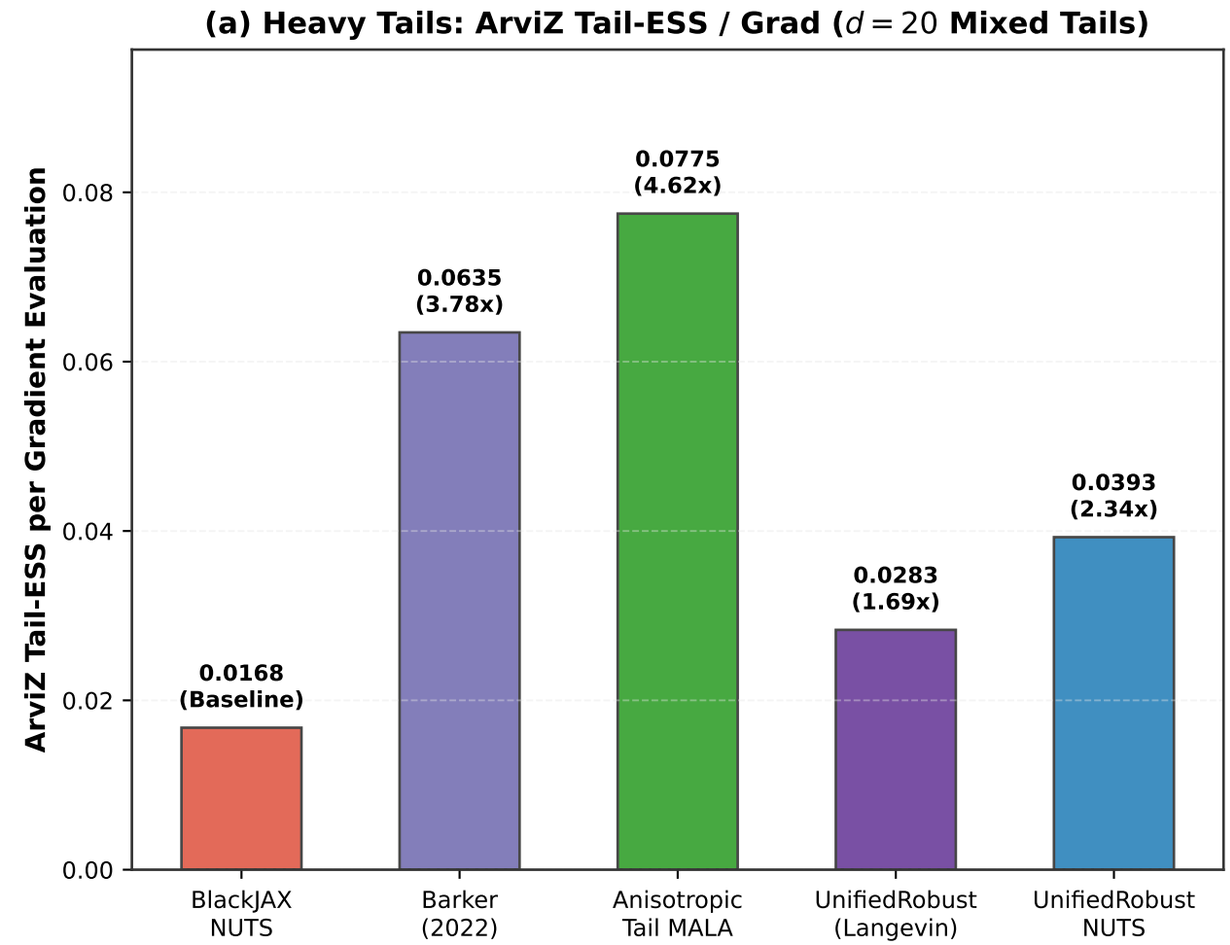
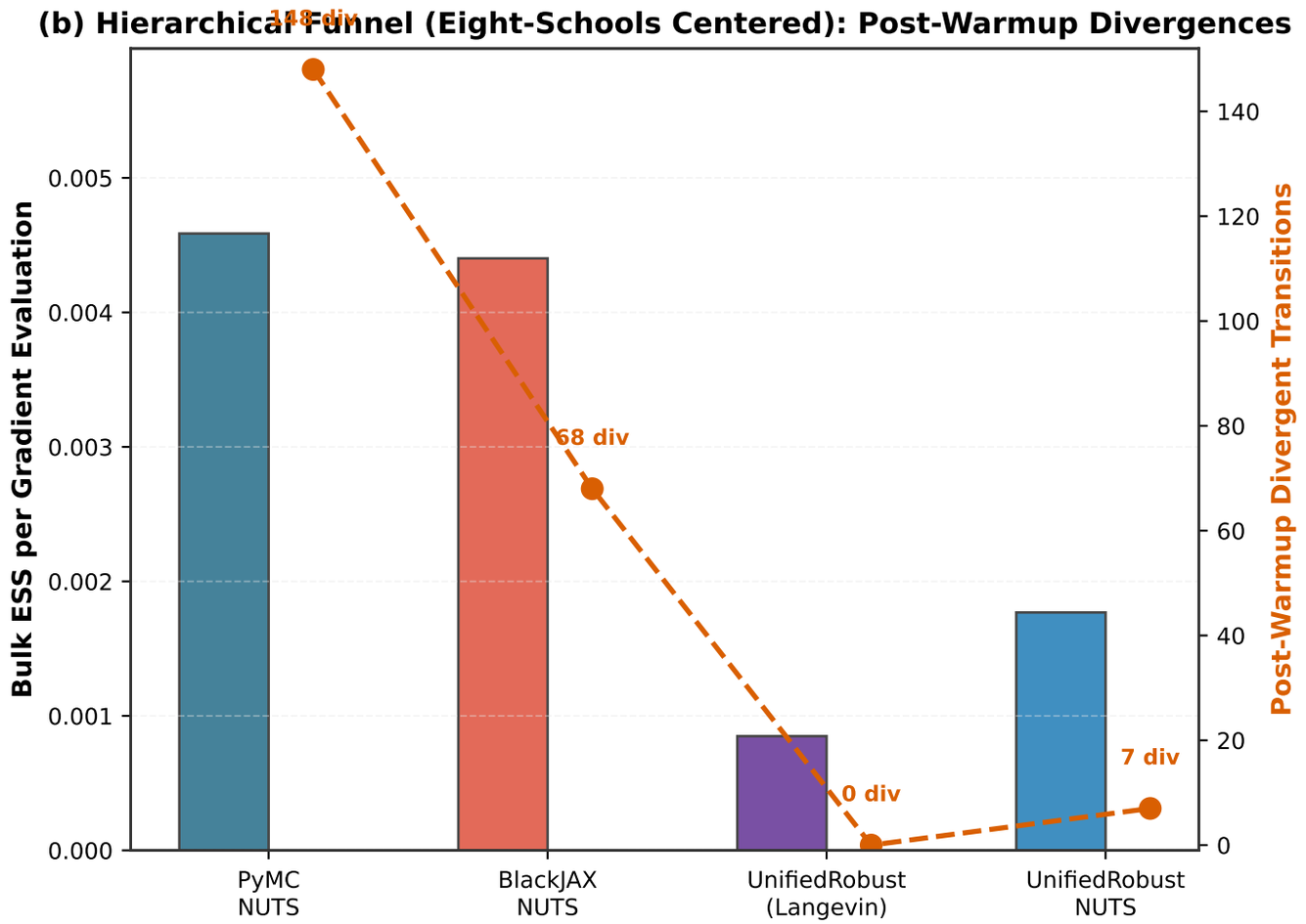
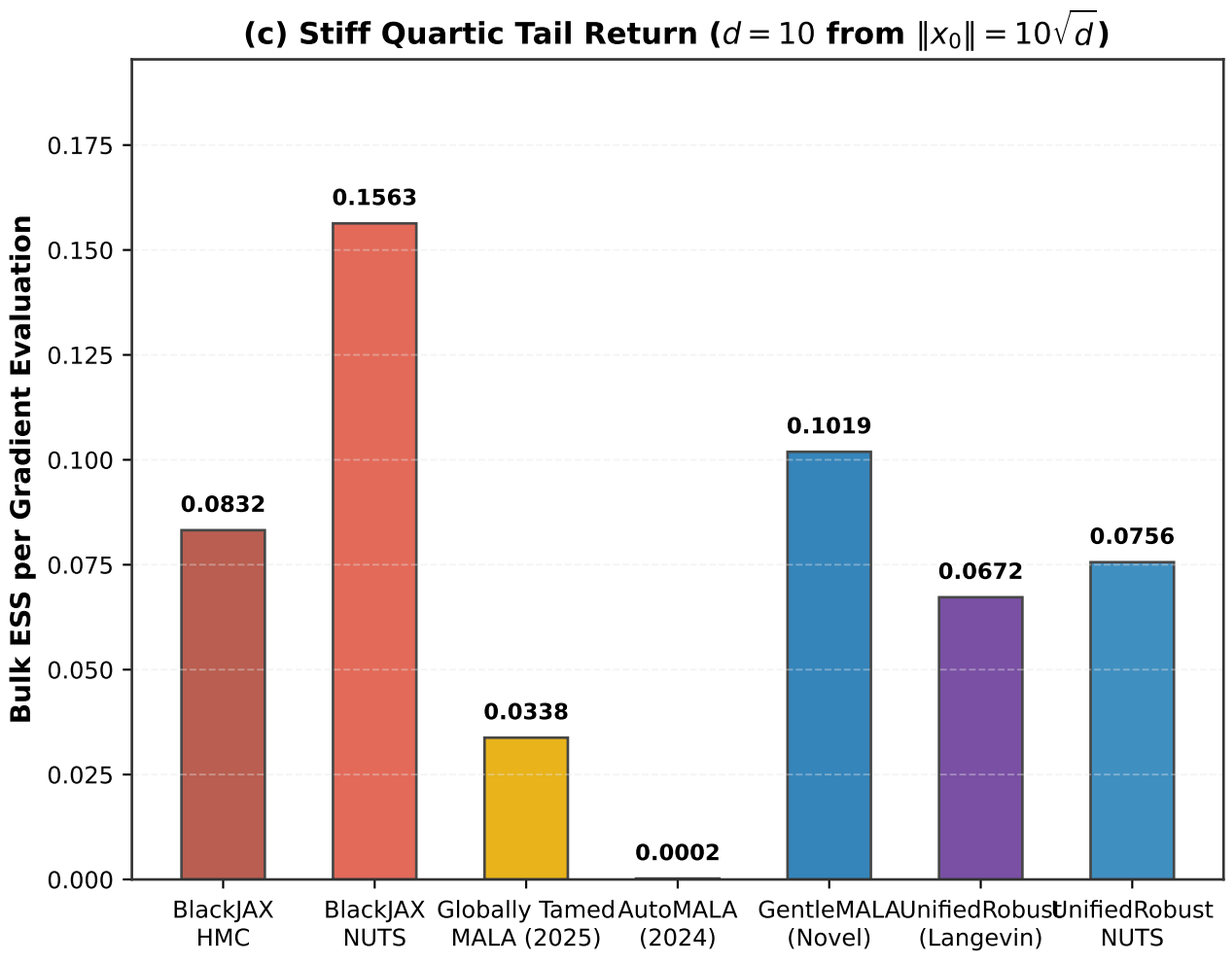
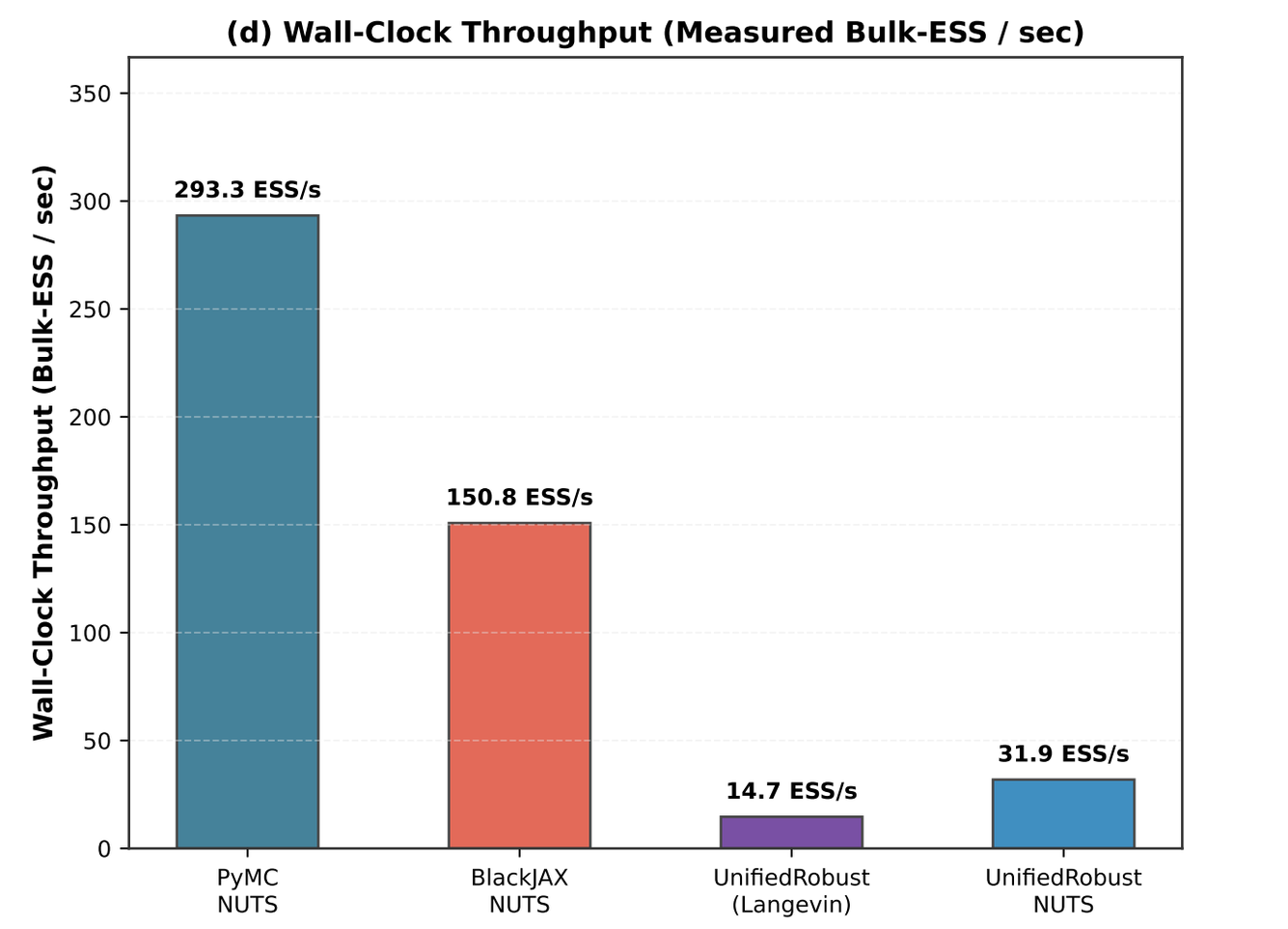

(a) Heavy Tails: ArviZ Tail-ESS / Grad ($d = 20$ Mixed Tails)
ArviZ Tail-ESS per Gradient Evaluation
0.00
0.02
0.04
0.06
0.08
0.0168
(Baseline)
0.0635
(3.78x)
0.0775
(4.62x)
0.0283
(1.69x)
0.0393
(2.34x)
BlackJAX
NUTS
Barker
(2022)
Anisotropic
Tail MALA
UnifiedRobust
(Langevin)
UnifiedRobust
NUTS


(b) Hierarchical Funnel (Eight-Schools Centered): Post-Warmup Divergences
Bulk ESS per Gradient Evaluation
Post-Warmup Divergent Transitions
0.000
0.001
0.002
0.003
0.004
0.005
0
20
40
60
80
100
120
140
148 div
68 div
0 div
7 div
PyMC
NUTS
BlackJAX
NUTS
UnifiedRobust
(Langevin)
UnifiedRobust
NUTS


(c) Stiff Quartic Tail Return ($d = 10$ from $\|x_0\| = 10\sqrt{d}$)
Bulk ESS per Gradient Evaluation
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.0832
0.1563
0.0338
0.0002
0.1019
0.0672
0.0756
BlackJAX
HMC
BlackJAX
NUTS
Globally Tamed
MALA (2025)
AutoMALA
(2024)
GentleMALA
(Novel)
UnifiedRobust
(Langevin)
UnifiedRobust
NUTS


(d) Wall-Clock Throughput (Measured Bulk-ESS / sec)
Wall-Clock Throughput (Bulk-ESS / sec)
0
50
100
150
200
250
300
350
293.3 ESS/s
150.8 ESS/s
14.7 ESS/s
31.9 ESS/s
PyMC
NUTS
BlackJAX
NUTS
UnifiedRobust
(Langevin)
UnifiedRobust
NUTS